MPI - IMP PYQ Analysis

Microprocessor and Interfacing | BE05000551 | Sem 5 | GTU

New Syllabus 2024-25 | Papers Analyzed: 12 (S20-W25)

> **Compiled BY** : Manthan A. D. [in]

📌 M.IMP Questions [High → Low Priority]

1. **8085 Internal Architecture / Functional Block Diagram** (8085 Block Diagram)

   > (**Ch.** 3) | **Freq:** 10 | **Max Marks:** 7 | [Theory: 2] [Diag: 8]

📌 *Most Asked Across ALL 12 papers. Non-negotiable. Draw it cold.*

2. **8085 Flag Register - All 5 Bits (S, Z, AC, P, CY) with Set/Reset Conditions** (Flag Register)

   > (**Ch.** 3) | **Freq:** 9 | **Max Marks:** 7 | [Theory: 5] [Def: 4]

📌 *Previous doc merged this with Programming Model and undercounted. Appears standalone 9 times. Highest-frequency non-diagram topic.*

3. **8085 Interrupts - All Types (HW+SW), Vector Locations, Maskability, Priority** (8085 Interrupts Full)

   > (**Ch.** 7) | **Freq:** 9 | **Max Marks:** 7 | [Theory: 3] [Def: 4]

📌 *Appears in 9 papers. Previous doc had freq 7 - undercounted SumVI2023, WinVI2023, WinVI2024.*

4. **Addressing Modes of 8085 - All Types with Examples (Register, Immediate, Direct, Indirect, Implied)** (Addressing Modes)

   > (**Ch.** 4) | **Freq:** 8 | **Max Marks:** 7 | [Theory: 4] [Def: 4]

5. **8255A Programmable Peripheral Interface - Block Diagram, Modes (0/1/2), BSR Mode, Control Word Format** (8255A Block+Modes)

   > (**Ch.** 7) | **Freq:** 8 | **Max Marks:** 7 | [Theory: 3] [Diag: 5]

6. **Memory-Mapped I/O vs Peripheral (I/O-Mapped) I/O - Compare** (MM I/O vs PM I/O)

   > (**Ch.** 2) | **Freq:** 7 | **Max Marks:** 7 | [Theory: 4] [Def: 3]

📌 *Previous doc capped max at 4. WinIV2025 has it in a 7-mark combined question - max corrected to 7.*

7. **Timing Diagram - Any 1-Byte Instruction (MOV C,A / MOV A,B / MOV A,C / MOV M,D); Define T-State, Machine Cycle, Instruction Cycle** (Timing Diag 1-Byte)

   > (**Ch.** 3) | **Freq:** 7 | **Max Marks:** 7 | [Diag: 7]

📌 *SumIV2025 and WinIV2025 both added new instances - more recent than previous doc.*

8. **8259A Programmable Interrupt Controller - Internal Block Diagram, Functions of Each Block** (8259A Block)

   > (**Ch.** 7) | **Freq:** 7 | **Max Marks:** 7 | [Theory: 2] [Diag: 5]

📌 *WinVI2024 has a 4-mark version too - confirms sustained testing.*

9. **ALP - Ascending / Descending Sort of N Numbers (Bubble Sort)** (Sort ALP)

   > (**Ch.** 5) | **Freq:** 7 | **Max Marks:** 7 | [Num: 7]

📌 *Both ascending and descending asked. Learn both variants of the same algorithm.*

10. **8085 Programming Model - Register Set (B,C,D,E,H,L,A,SP,PC,IR)** (Prog Model)

    > (**Ch.** 3) | **Freq:** 7 | **Max Marks:** 7 | [Theory: 3] [Diag: 2] [Def: 2]

📌 *Previous doc merged this with Flag Register. Both appear combined AND standalone - keep separate, both M.IMP.*

11. **RIM and SIM Instructions - Describe Each Bit in Detail** (RIM/SIM)

    > (**Ch.** 7) | **Freq:** 7 | **Max Marks:** 7 | [Theory: 2] [Def: 4]

📌 *Previous doc had max as 4. Corrected: WinIV2020 Q5c is explicitly 7 marks for full RIM+SIM.*

12. **Timing Diagram - Any 2-Byte Instruction (MVI A,45H / MVI B,22H / MVI D,88H); Define T-State, Machine Cycle, Instruction Cycle** (Timing Diag 2-Byte)

    > (**Ch.** 3) | **Freq:** 6 | **Max Marks:** 7 | [Diag: 6]

📌 *WinIV2025 alone has both a 1-byte AND 2-byte timing diagram question. Always prep both.*

13. **ALP - Find Largest / Smallest Number from Array of N Data Bytes; Store Result** (Max/Min ALP)

    > (**Ch.** 5) | **Freq:** 6 | **Max Marks:** 7 | [Num: 6]

14. **EPROM Interfacing with 8085 using 3-to-8 Decoder (74LS138); Show Address Decoding + Memory Map** (EPROM Interface)

    > (**Ch.** 3) | **Freq:** 6 | **Max Marks:** 7 | [Diag: 6]

📌 *Includes 4KB EPROM + RAM combined interfacing variations. Learn address map derivation.*

15. **Demultiplexing of Low-Order Address/Data Bus (AD0-AD7) - ALE + Latch Circuit Diagram** (Demux Bus)

    > (**Ch.** 3) | **Freq:** 6 | **Max Marks:** 7 | [Theory: 1] [Diag: 4]

📌 *Appears as 3-mark, 4-mark, and WinIV2025 has it embedded in a 7-mark question.*

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16. **8085 Pin Diagram - All Pins with Functions incl. Control and Status Signals** (Pin Diagram)

    > (**Ch.** 3) | **Freq:** 5 | **Max Marks:** 7 | [Diag: 5]

17. **8085 Pin Functions - ALE, INTR, HOLD, READY, HLDA, SOD, SID, IO/M, TRAP (4-mark slot)** (Pin Functions Short)

    > (**Ch.** 3) | **Freq:** 5 | **Max Marks:** 7 | [Theory: 3] [Def: 2]

📌 *Separate from full Pin Diagram. Often asked as short 3-4 mark standalone questions.*

18. **Timing Diagram - ADD M / MOV A,B (78H); Name Machine Cycles, Draw T-State Diagram** (Timing Diag Misc)

    > (**Ch.** 3) | **Freq:** 5 | **Max Marks:** 7 | [Diag: 5]

📌 *Memory-reference instructions like ADD M involve 3 machine cycles. Different from standard MOV.*

19. **8086 Logical Block Diagram - BIU, EU, Segmentation, Architecture** (8086 Block Diagram)

    > (**Ch.** 8) | **Freq:** 5 | **Max Marks:** 7 | [Theory: 3] [Diag: 4]

📌 *SumVI2023 asks both architecture AND pin diagram together in 7 marks. Prep together.*

20. **80386 Microprocessor - Architecture, Programming Model, Data Types, Segments** (80386 Arch)

    > (**Ch.** 8) | **Freq:** 5 | **Max Marks:** 7 | [Theory: 4] [Diag: 2]

21. **80286 Microprocessor - Architecture, Block Diagram, Programming Model, Protected Mode** (80286 Arch)

    > (**Ch.** 8) | **Freq:** 5 | **Max Marks:** 7 | [Theory: 2] [Diag: 3]

22. **Instruction Cycle, Machine Cycle, T-State - Definitions + Identify Machine Cycles for Given Instructions** (Instruction/Machine Cycles)

> (**Ch.** 3) | **Freq:** 4 | **Max Marks:** 7 | [Theory: 4]

23. **ALP - Block Transfer (Copy N Bytes from One Memory Location to Another)** (Block Transfer ALP)

    > (**Ch.** 5) | **Freq:** 4 | **Max Marks:** 7 | [Num: 4]

24. **CALL and RET Instructions - Subroutine Concept with Examples** (CALL + RET)

    > (**Ch.** 6) | **Freq:** 4 | **Max Marks:** 7 | [Theory: 3] [Num: 1]

25. **Delay Loop Design - Generate Specific Delay (200ms / 1ms / 1sec); Show Calculation for Given Clock Frequency** (Delay Loop ALP)

    > (**Ch.** 6) | **Freq:** 4 | **Max Marks:** 7 | [Theory: 1] [Num: 4]

26. **Stack Concept + Stack Pointer; PUSH and POP Instructions with Examples** (Stack + PUSH/POP)

    > (**Ch.** 6) | **Freq:** 6 | **Max Marks:** 4 | [Theory: 3] [Num: 1] [Def: 2]

📌 *Never asked in 7-mark but Freq 6 across both semesters. Classic 3-4 mark slot staple.*

27. **MEMR / MEMW / IOR / IOW Control Signal Generation in 8085 - Circuit Diagram** (Control Signal Gen)

    > (**Ch.** 3) | **Freq:** 4 | **Max Marks:** 4 | [Theory: 1] [Diag: 3]

28. **8085 Bus Organization - Data Bus, Address Bus, Control Bus (Structure + Width)** (Bus Organization)

    > (**Ch.** 3) | **Freq:** 4 | **Max Marks:** 4 | [Theory: 3] [Def: 1]

29. **Interrupt Classification - Maskable vs Non-Maskable, Vectored vs Non-Vectored, HW vs SW** (Interrupt Classification)

    > (**Ch.** 7) | **Freq:** 4 | **Max Marks:** 4 | [Theory: 4] [Def: 1]

30. **8086 Segmentation - Segment Registers (CS, DS, SS, ES), Physical Address Calculation** (8086 Segmentation)

    > (**Ch.** 8) | **Freq:** 4 | **Max Marks:** 4 | [Theory: 3] [Def: 1]

31. **8086 Minimum Mode vs Maximum Mode - Compare** (8086 Min/Max Mode)

    > (**Ch.** 8) | **Freq:** 4 | **Max Marks:** 4 | [Theory: 4]

32. **ALP - Hexadecimal / Binary Counter (Count 00H to FFH or FFH to 00H) with Delay; Display at Output Port** (Hex Counter ALP)

    > (**Ch.** 6) | **Freq:** 3 | **Max Marks:** 7 | [Num: 3]

33. **BCD Number to Hexadecimal (Binary) Conversion ALP; Store Result** (BCD to Hex ALP)

    > (**Ch.** 6) | **Freq:** 3 | **Max Marks:** 4 | [Num: 3]

🟡 IMP Questions [High → Low Priority]

1. **Instruction Format of 8085 - One-Byte, Two-Byte, Three-Byte with Examples and Classification** (Instruction Format)

    > (**Ch.** 4) | **Freq:** 3 | **Max Marks:** 7 | [Theory: 3] [Def: 1]

2. **ALP - Add Two Numbers (B+C); If Sum > FFH Display 01H at Port; Else Display Sum** (Carry Check ALP)

    > (**Ch.** 5) | **Freq:** 3 | **Max Marks:** 4 | [Num: 3]

📌 *Classic conditional check ALP pattern. Previous doc counted freq 3 correctly.*

3. **Arithmetic Instructions of 8085 - ADD, SUB, INR, DCR, ADI, SBI, DAA with Examples** (Arithmetic Instructions)

    > (**Ch.** 4) | **Freq:** 4 | **Max Marks:** 4 | [Theory: 3] [Def: 1]

4. **8086 Segment Registers (CS, DS, SS, ES) - List and Explain** (8086 Seg Registers)

> (**Ch.** 8) | **Freq:** 3 | **Max Marks:** 4 | [Def: 3]

5. **ALP - Find 2's Complement of a Number; Store Result at Memory** (2's Complement ALP)

   > (**Ch.** 5) | **Freq:** 2 | **Max Marks:** 4 | [Num: 2]

6. **Counters and Time Delay Routines - Explain with Example; List Applications** (Counters + Time Delay Theory)

   > (**Ch.** 6) | **Freq:** 2 | **Max Marks:** 7 | [Theory: 2] [Num: 1]

7. **Differentiate 8085 vs 8086 Microprocessor** (8085 vs 8086)

   > (**Ch.** 8) | **Freq:** 3 | **Max Marks:** 3 | [Def: 3]

8. **8255A BSR Mode - Explain in Detail** (BSR Mode 8255A)

   > (**Ch.** 7) | **Freq:** 2 | **Max Marks:** 4 | [Theory: 2]

📌 *Appears as sub-part of M.IMP 8255A entry but also tested standalone. Know separately.*

9. **ALP - Nibble Separation + Convert Each Nibble to ASCII Hex Code; Store in Memory** (Nibble ASCII ALP)

   > (**Ch.** 6) | **Freq:** 2 | **Max Marks:** 7 | [Num: 2]

10. **Absolute Decoding vs Partial Decoding - Compare; Foldback Memory Space** (Abs vs Partial Decoding)

    > (**Ch.** 3) | **Freq:** 3 | **Max Marks:** 3 | [Theory: 2] [Def: 1]

11. **Differentiate Assembly Language vs Machine Language (or High-Level Language) - With Examples** (Asm vs ML)

    > (**Ch.** 1) | **Freq:** 3 | **Max Marks:** 3 | [Theory: 2] [Def: 1]

12. **80286/80386 Segment Descriptor, Descriptor Table (GDT/LDT), and Selectors** (Descriptor Table)

    > (**Ch.** 8) | **Freq:** 3 | **Max Marks:** 4 | [Theory: 2] [Def: 1]

13. **PUSH/POP vs CALL/RET - Compare; Stack Instructions vs Subroutine Instructions** (PUSH/POP vs CALL/RET)

    > (**Ch.** 6) | **Freq:** 2 | **Max Marks:** 3 | [Theory: 3]

14. **Subroutine Nesting + Conditional CALL Instructions** (Nesting Subroutines)

    > (**Ch.** 6) | **Freq:** 2 | **Max Marks:** 3 | [Theory: 2]

15. **8259A Initialization - ALP Program to Initialize 8259 PIC Connected to 8085** (8259A Init ALP)

    > (**Ch.** 7) | **Freq:** 2 | **Max Marks:** 7 | [Num: 2]

16. **MOV C,A Data Flow - Fetch-Decode-Execute Steps Inside MPU** (MOV C,A Execution Steps)

    > (**Ch.** 3) | **Freq:** 2 | **Max Marks:** 7 | [Theory: 2]

17. **Specific 8085 Instructions with Byte Count, Machine Cycles, T-States - SHLD, RAL, CALL, CPI, LHLD, MVI M** (Instruction Details)

    > (**Ch.** 4) | **Freq:** 2 | **Max Marks:** 4 | [Theory: 2]

18. **RLC vs RAL; RRC vs RAR - Difference with Flag Effects** (Rotate Instructions Diff)

    > (**Ch.** 4) | **Freq:** 2 | **Max Marks:** 3 | [Theory: 2]

19. **ALP - Descending Order Sort (N Numbers)** (Descending Sort ALP)

    > (**Ch.** 5) | **Freq:** 2 | **Max Marks:** 7 | [Num: 2]

📌 *Sub-variant of Sort ALP (M.IMP #9), but GTU asks it separately by name often enough.*

20. **ALP - Separate Positive and Negative Values from Array (Signed Number Handling)** (Sign Separation ALP)

    > (**Ch.** 5) | **Freq:** 2 | **Max Marks:** 7 | [Num: 2]

21. **Memory Interfacing with 8085 - Key Concepts + RAM Interface Circuit** (RAM Interfacing)

    > (**Ch.** 3) | **Freq:** 2 | **Max Marks:** 7 | [Theory: 1] [Diag: 1]

22. **I/O Device Interfacing with 8085 - Steps + Block Diagram of I/O Interface Requirements** (I/O Device Interface)

    > (**Ch.** 7) | **Freq:** 2 | **Max Marks:** 7 | [Theory: 1] [Diag: 1]

23. **ALP - Binary/Hex to Unpacked BCD Conversion (Store Each BCD Digit Separately)** (BCD Conversion ALP)

    > (**Ch.** 6) | **Freq:** 2 | **Max Marks:** 7 | [Num: 2]

New Syllabus Blind Spots [Zero PYQ Coverage]

Study Anyway

- Components of a Microprocessor - Internal structure of ALU and Control & Timing Unit specifically (Ch. 1)
- Microprocessor Systems with Bus Organization - System-level view: CPU + Memory + I/O on shared bus (Ch. 1)
- Microprocessor Architecture and Operations - General microprocessor architecture distinctions, not 8085-specific (Ch. 2)
- Memory and I/O Operations - Read/write cycle at the abstract, non-8085-specific level (Ch. 2)
- I/O Devices - Types and classification of I/O devices in a microprocessor system (Ch. 2)
- Writing, Assembling and Executing a Program / Debugging - Assembler directives, linker steps, debugger usage (Ch. 4)
- Instruction and Data Formats - Bit-level encoding of 8085 instructions: opcode field, register field, data field (Ch. 4)
- 16-Bit Data Operations - LXI, DAD, DCX, INX used specifically for 16-bit arithmetic programs (Ch. 6)

- BCD Arithmetic - DAA instruction in multi-byte BCD addition/subtraction programs (Ch. 6)
- Interfacing Concepts - Port types at conceptual level; memory-mapped port vs I/O-mapped port distinction (Ch. 7)
- 8086 Pin Functions - Detailed explanation of all 8086 pins (AD0-AD15, BHE, S0-S2, DEN, DT/R) in min/max mode (Ch. 8)
- 80386 Instruction Set Specifics - Extended 32-bit instruction differences from 8086 (Ch. 8)
- Segment Types in 80286/80386 - Code, data, stack, system segment details and their attributes (Ch. 8)
- Descriptor Table - GDT, LDT, IDT structure; selector field encoding in 80286/80386 (Ch. 8)

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